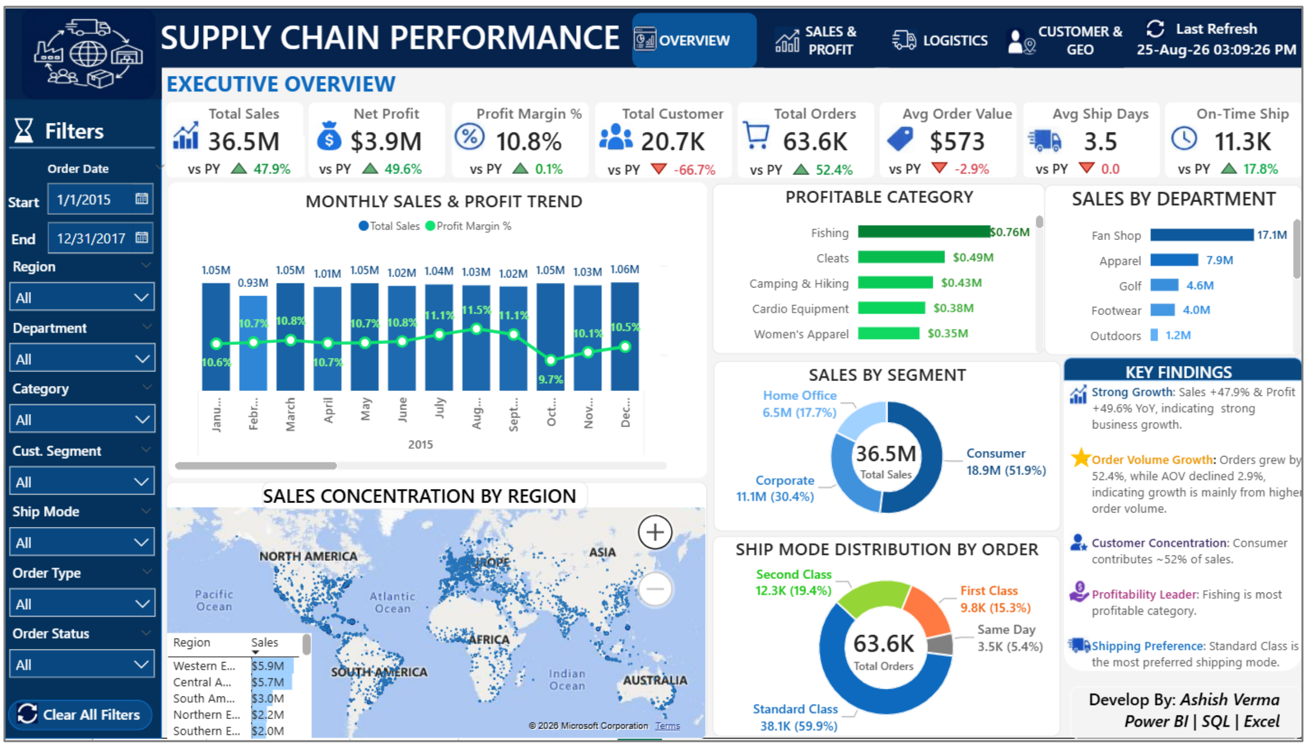
SUPPLY CHAIN PERFORMANCE ANALYSIS

End-to-End Data Analytics & MIS Project Report



Excel Power Query → Feature Engineering → Power BI Data Model & Dashboard → SQL Independent Validation

Dataset coverage: 01 Jan 2015 to 31 Jan 2018 • 180,519 line-item records • 43 fields

Prepared from the Supply_Chain_Cleaned_Data.xlsx, Supply_Chain_Performance_Analysis.pbix, Supply_Chain_SQL_Validation.sql and the five supplied image-evidence files.

1. Executive Summary

This project delivers a complete supply-chain analytics workflow from structured data preparation through interactive Power BI reporting and independent SQL validation. The supplied artifacts show a single operational dataset being transformed in Excel Power Query, modeled as a dimensional Power BI semantic model, presented through four dashboard pages, and validated through a dedicated MySQL analysis script.

The cleaned dataset contains 180,519 records across 43 columns, covering 01 Jan 2015 through 31 Jan 2018. It represents 180,519 order-item records, 65,752 distinct orders, 20,652 customers, 118 products, 50 categories, 164 countries, 23 regions, 1089 states and 11 departments.

Project layer	Primary tool	Purpose	Output
Data preparation	Excel + Power Query	Extract, clean, standardize and engineer analytical fields	Cleaned 43-column dataset
Semantic modeling	Power BI	Build reusable dimensions, fact table and measures	Star-style analytical model
Visualization	Power BI	KPI monitoring, trends, profitability, logistics, customer and geography analysis	4-page interactive dashboard
Independent validation	MySQL / SQL	Recalculate business metrics independently of Power BI	SQL validation and data-quality checks

Management-level outcome

- Overall dataset totals are approximately $36.8M sales and $4.0M net profit, with an aggregate profit margin of 10.8%.
- Consumer is the largest customer segment by sales ($19.1M), followed by Corporate and Home Office.
- Fan Shop is the largest department by sales ($17.1M), while Fishing is the leading category by sales ($6.9M).
- Standard Class is the dominant shipping mode (39.3K distinct orders), making logistics reliability in this mode especially important.
- Western Europe and Central America are the two largest regions by sales in the supplied data.

Important interpretation note: dashboard screenshots supplied with the project contain different date selections on different pages. Therefore, the report distinguishes between full-dataset calculations and the 2015–2017 calculations used for the main overview validation rather than treating every screenshot KPI as if it used one identical filter state.

2. Report Structure

This report documents the project as a standalone analytical deliverable. The sections follow the actual implementation sequence and then move from dashboard interpretation to independent SQL validation.

Section	Purpose
1. Executive Summary	Project documentation
2. Report Structure	Project documentation
3. Project Scope, Objectives & Business Questions	Project documentation
4. Data Source & Dataset Profiling	Project documentation
5. End-to-End Analytical Workflow	Project documentation
6. Excel Power Query — Data Cleaning & Feature Engineering	Project documentation
7. Power BI Data Model & Semantic Layer	Project documentation
8. Power BI Dashboard — Page-by-Page Documentation	Project documentation
9. Dashboard Evidence & Visual Interpretation	Project documentation
10. . SQL Layer – Independent Validation Framework	Project documentation
11. Independent Validation Results	Project documentation
12. Business Findings	Project documentation
13. Business Recommendations	Project documentation
14. Technical Implementation Notes	Project documentation
Appendix A	Power BI Measures Catalog
Appendix B	SQL-to-Dashboard Validation Map
Appendix C	Scope, Assumptions & Limitations
15. Detailed Data Preparation Logic	Project documentation
16. Power BI Data Model & Measures	Project documentation
17. SQL Validation & Final Business Insights	Project documentation

3. Project Scope, Objectives & Business Questions

3.1 Business objective

The SQL file defines the business objective as measuring supply-chain commercial performance, profitability, customer contribution, product/category performance, shipping reliability, departmental performance, regional performance and year-over-year trends. The Power BI report operationalizes those objectives into a multi-page management dashboard.

3.2 Core business questions

1. How much sales and profit are being generated, and what is the overall profit margin?
2. How are sales, profit and order volumes trending over time?
3. Which categories, products and departments contribute the most commercial value?
4. Which products create the greatest loss exposure?
5. Which customer segments and customers contribute the most sales and profit?
6. How reliable are shipments by mode, status and region?
7. Where are the strongest geographic markets and concentrations?
8. How are key KPIs changing year over year?
9. Are the Power BI metrics independently reproducible in SQL?
10. Does the cleaned dataset satisfy critical data-quality rules?

3.3 Intended users

- MIS / reporting teams
- Business and operations managers
- Sales and commercial teams
- Supply-chain and logistics teams
- Management / leadership
- Data analysts performing independent validation

3.4 Analytical grain

The source table is a line-item level dataset: an Order_ID can occur on multiple rows because one order can contain multiple order items/products. The supplied SQL explicitly treats the analysis as a single-table analysis and does not require joins.

4. Data Source & Dataset Profiling

Profile item	Result
Cleaned data file	Supply_Chain_Cleaned_Data.xlsx
Excel working file	Supply_Chain_Cleaned_Data.xlsx
Power BI model	Supply_Chain_Performance_Analysis.pbix
SQL validation file	Supply_Chain_SQL_Validation.sql
Process/dashboard evidence	05_Screenshots (Power Query + Power BI dashboard evidence)
Rows in cleaned CSV	180,519
Columns	43
Distinct order items	180,519
Distinct orders	65,752
Distinct customers	20,652
Distinct products	118
Distinct categories	50
Distinct countries	164
Distinct regions	23
Distinct states	1089
Distinct departments	11
Order-date range	01 Jan 2015 – 31 Jan 2018

Database source: Kaggle — DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS | https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

4.1 Field domains represented

Domain	Representative fields
Order & transaction	Order_ID, Order_Date, Order_Type, Order_Item_ID, Order_Quantity
Product	Product_ID, Product_Category_ID, Category_Name, Product_Name
Customer	Customer_ID, Customer_Name, Customer_Segment
Commercial	Price_Per_Product, Order_Item_Discount, Discount_Rate, Sales, Order_Profit, Profit_Margin_Percent
Profit classification	Discount_Range, Profit_Category
Shipping	Ship_Date, Ship_Mode, Scheduled_Ship_Days, Actual_Ship_Days, Ship_Delay_Days, Ship_Status
Order lifecycle	Order_Status, Late_Delivery_Risk
Geography	Market, Order_Region, Order_Country, Order_State, Order_City, Latitude, Longitude
Organization	Department_ID, Department_Name

4.2 Source workbook structure

Inspection of the supplied workbook shows three sheets: Table1, Sheet1 and DataCoSupplyChainDataset. Table1 contains the 43-field cleaned analytical structure with 180,520 worksheet rows including the header, corresponding to 180,519 data records in the cleaned CSV. The workbook also contains an Excel query connection named “Query - Table1”, indicating the cleaned Table1 output is query-driven.

5. End-to-End Analytical Workflow

5.1 Overall architecture

The project follows the workflow requested in the supplied project artifacts:

Source / workbook → Excel Power Query extraction → data cleaning → feature engineering → cleaned CSV / Excel output → Power BI model → DAX measures → four-page dashboard → MySQL import → independent SQL validation → business findings

5.2 Why the layered approach matters

- Excel Power Query provides a transparent and repeatable preparation layer.
- Feature engineering moves reusable business classifications into the analytical dataset.
- Power BI separates the reusable semantic model from the presentation layer.
- DAX measures provide interactive calculations under slicer/filter context.
- SQL validation creates an independent calculation path and reduces the risk of validating a report only against itself.

5.3 Deliverables produced

Deliverable	Role in project
Cleaned Excel/CSV	Prepared analytical source with standardized fields and engineered features
Power BI PBIX	Dimensional model, measures, navigation, filters, KPI cards and visual analytics
SQL script	Database/table setup, import, KPI calculations, trend analysis and quality validation
Dashboard/screenshots	05_Screenshots: Power Query applied steps and four Power BI dashboard pages
Final report	Professional documentation of the complete analytical workflow and findings

6. Excel Power Query — Data Cleaning & Feature Engineering

The supplied image evidence contains a Power Query Editor screenshot for a query named “Cleaned_Data”. The visible Applied Steps are preserved below as the source-derived preparation sequence.

6.1 Applied Steps captured from the supplied Power Query screenshot

01. Source

02. Reordered Columns 1

03. Changed Type

04. Replaced 'and' with & in dept_name

05. Removed constant&unnecessary Columns

06. Replaced _ in order status

07. Replaced Shipping Cancelled

08. Capitalized Each Word

09. Changed Type with Locale

10. Changed Type 1

11. Renamed Columns

12. Added Ship_Delay_Days

13. Added Profit_Category

14. Concatenate Customer_Name_Surname

15. Added Discount_Range

16. Removed Columns2

17. Changed Type2

18. Reordered Columns2

19. Removed Duplicates

6.2 Preparation logic represented by the steps

Transformation area	Evidence from applied steps	Analytical purpose
Column organization	Reordered Columns 1 / 2	Create a usable and stable analytical column order
Data types	Changed Type / Changed Type with Locale / Changed Type 1 / Changed Type2	Ensure dates, numeric measures and text fields are interpreted consistently
Text standardization	Replaced 'and' with &, Replaced _, Capitalized Each Word, Renamed Columns	Improve consistency for categories, department and status labels
Noise removal	Removed constant&unnecessary Columns / Removed Columns2	Reduce irrelevant source fields
Shipping feature	Added Ship_Delay_Days	Create a derived operational timing metric
Profit feature	Added Profit_Category	Classify profitability for business analysis
Customer feature	Concatenate Customer_Name_Surname	Create a standardized customer display/name field
Discount feature	Added Discount_Range	Create discount bands for impact analysis
Deduplication	Removed Duplicates	Protect the analytical dataset from duplicate records

6.3 Resulting analytical dataset

The final cleaned CSV contains 180,519 rows and 43 columns. There are no missing values in the six critical fields tested by the supplied SQL logic (Order_ID, Customer_ID, Product_ID, Order_Date, Sales and Order_Profit). Only Customer_ZIP_Code contains missing values, with 3 records.

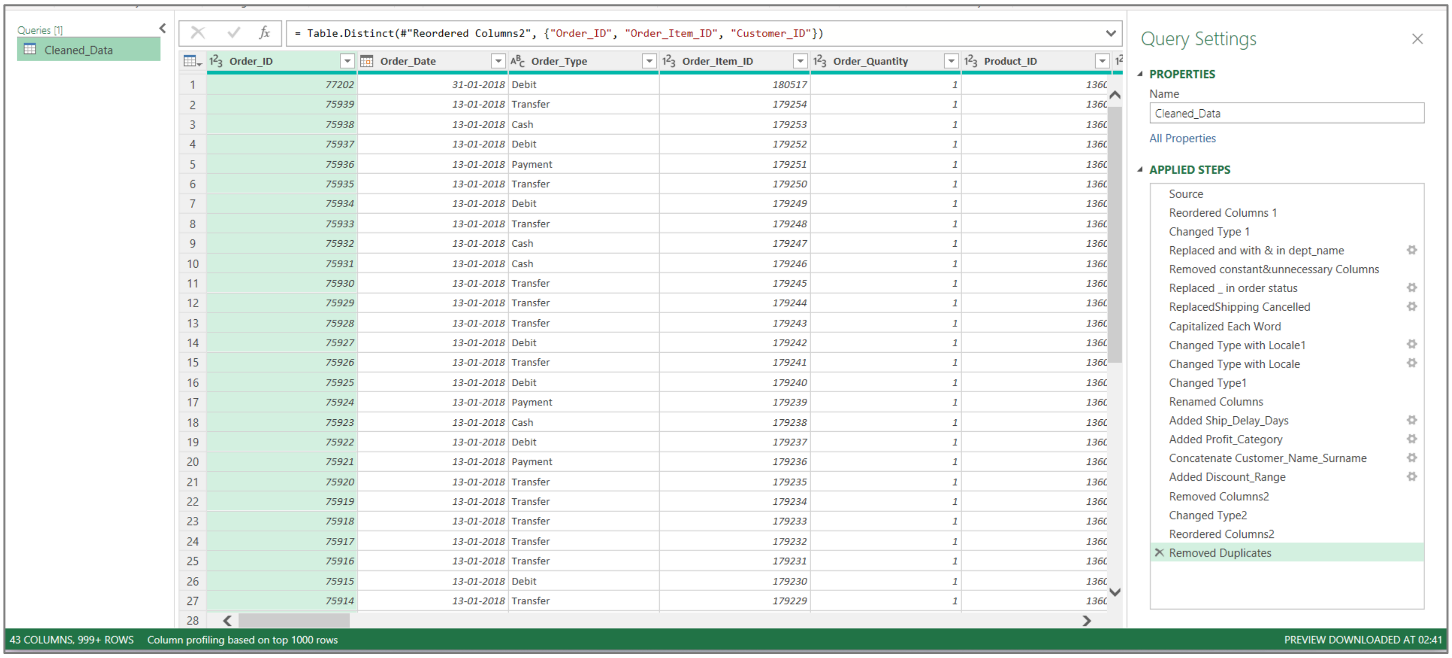


Figure 1. 1_Power_Quey_Applied_Steps.png — Power Query Editor evidence showing the Cleaned_Data query and its Applied Steps.

7. Power BI Data Model & Semantic Layer

Inspection of the supplied PBIX package shows a dimensional model with one central Fact_Order table and supporting dimensions. The report layout references the following model objects.

Model object	Role
Fact_Order	Central transactional fact table
Dim_Date	Date filtering and time-based analysis
Dim_Customer	Customer and segment analysis
Dim_Product	Product and category analysis
Dim_Location	Country, region and city geography
Dim_Ship	Shipping-mode analysis
Dim_Department	Department analysis
_Measures	Dedicated measure table for reusable DAX measures
Last Refresh	Refresh-date display
Table	Supporting table used by YoY comparison visuals

7.1 Modeling approach

- The central fact table retains the transactional grain.
- Dimension tables provide descriptive slicing attributes.
- A dedicated _Measures table centralizes KPI calculations.
- Dim_Date supports date slicers and monthly/yearly trend analysis.
- Geography is separated into Dim_Location, allowing region/country/city analysis without duplicating all geographic attributes in every visual.

7.2 Measure framework identified in the PBIX

Core commercial

Total Sales • Net Profit • Profit Margin % • Total Orders • Total Customer • Avg Order Value

Profitability

Total Profitable Order • Total Loss • Avg Profit Per Cust • Avg Profit Per Order • Avg Discount %

Logistics

Total Order Delivered • Late Ship Order • Late Ship % • Avg Ship Days • Avg Scheduled Ship Days • Avg Ship Delay • Ship Cancelled Order • On-Time Ship Order • On Time Ship % • On-Time Ship Target

Coverage

Served Region • Served Countries • Served State • Sales Per Customer • Avg Sales Per Region • Avg Order Count Per Customer

YoY / comparison

YoY Sales % • YoY Profit % • YoY Profit Margin Change % • YoY Orders % • YoY Customers % • YoY Avg Order value % • YoY Avg Profit Per Cust % • YoY Avg Profit Per Order % • YoY Avg Ship Days • YoY Delay Days • YoY Discount% Change % • YoY Late Ship % % • YoY On Time % • YoY Order Delivered % • YoY Ship Cancelled % • YoY Total Loss % • YoY Total Profitable Orders %

8. Power BI Dashboard — Page-by-Page Documentation

8.1 Dashboard architecture

The PBIX contains four report pages: OVERVIEW, SALES & PROFIT, LOGISTICS and CUSTOMER & GEO. The page layout also contains common navigation, filter controls, refresh-date display and interactive visual components.

Page	Primary purpose	Major analytical content
OVERVIEW	Executive management view	Sales/profit KPIs, monthly trend, profitable categories, departments, segment mix, shipping mix, regional concentration and findings
SALES & PROFIT	Commercial performance	Profitability KPIs, monthly trend, discount impact, segment profit, category sales, product summary, department profit and findings
LOGISTICS	Operational reliability	Delivery KPIs, status by shipping mode, order status, late shipments by region, order/late trend, order type and on-time rate
CUSTOMER & GEO	Customer and geographic performance	Customer KPIs, segment performance, top customers, top countries, map, regional summary and findings

8.2 Common interactive controls

- Order Date
- Region
- Department
- Category
- Customer Segment
- Ship Mode
- Order Type
- Order Status

The PBIX layout shows these slicers across the dashboard pages, supporting cross-analysis without requiring separate static reports. A clear-all-slicers action is also present.

9. Dashboard Evidence & Visual Interpretation

9.1 Executive Overview

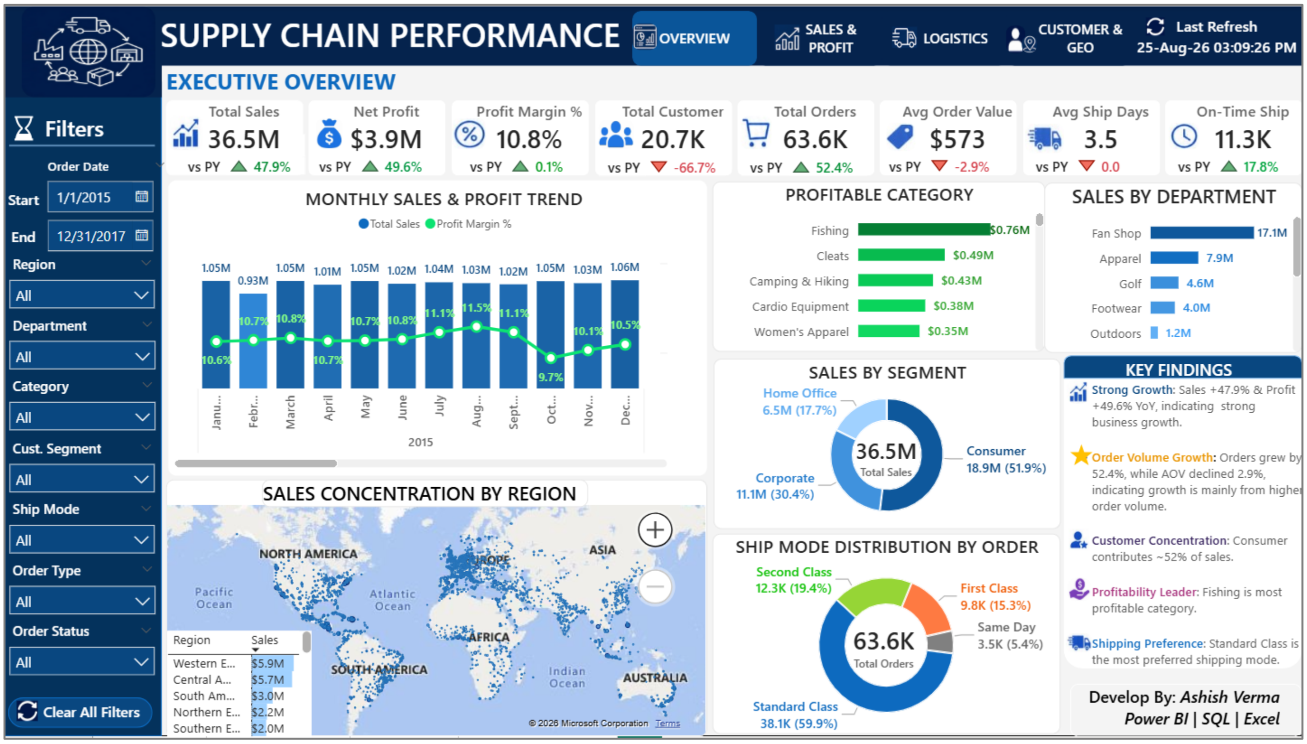


Figure 2. 01_Dashboard_Overview.png — OVERVIEW dashboard page. The screenshot shows KPI cards for sales, profit, margin, customers, orders, order value, shipping days and on-time shipments, followed by monthly trend, category, department, segment, regional and shipping visuals.

- The dashboard is designed as an executive-first page: headline KPIs appear above supporting diagnostic visuals.
- The monthly trend visual combines sales/profit with margin, making volume and profitability visible together.
- Sales by segment and shipping mode provide mix analysis, while the regional map adds geographic context.
- A Key Findings panel translates visual patterns into management-oriented statements.

What the page answers

- How much sales and net profit are being generated?
- What is the current profit margin and average order value?
- How are sales and profit changing over time?
- Which categories, departments and customer segments contribute most?
- How are orders distributed across shipping modes and regions?

The page is designed for management-level monitoring; detailed investigation is delegated to the Sales & Profit, Logistics and Customer & GEO pages.

9.2 Sales & Profit Analysis

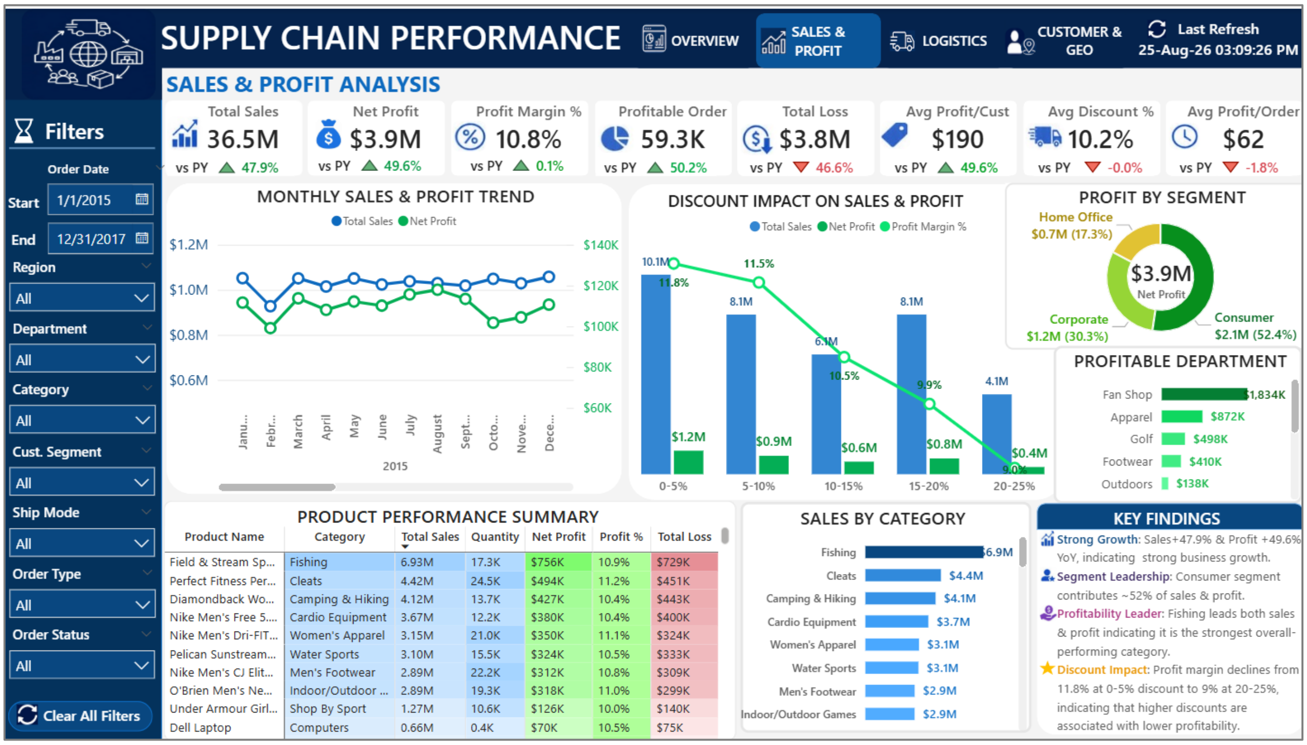


Figure 3. 02_Dashboard_Sales_Profit_Analysis.png — SALES & PROFIT dashboard page from the supplied project evidence.

The SALES & PROFIT page focuses on commercial performance and profitability rather than operational delivery.

- The discount-impact visual explicitly connects discount bands with sales, net profit and profit margin.
- The product performance summary combines sales, quantity, profit, margin and total loss, supporting product-level prioritization.
- Profit by segment and profitable department views help identify where commercial value is concentrated.

Key analytical views

- Sales, net profit, profit margin, total loss and average discount KPIs.
- Monthly sales and profit trend.
- Discount impact across discount ranges.
- Profit by customer segment.
- Sales by category.
- Product performance using sales, quantity, profit, margin and loss.
- Profit by department.

The product and discount views are particularly useful for identifying situations where high sales volume does not translate into healthy profitability.

9.3 Logistics & Delivery Performance

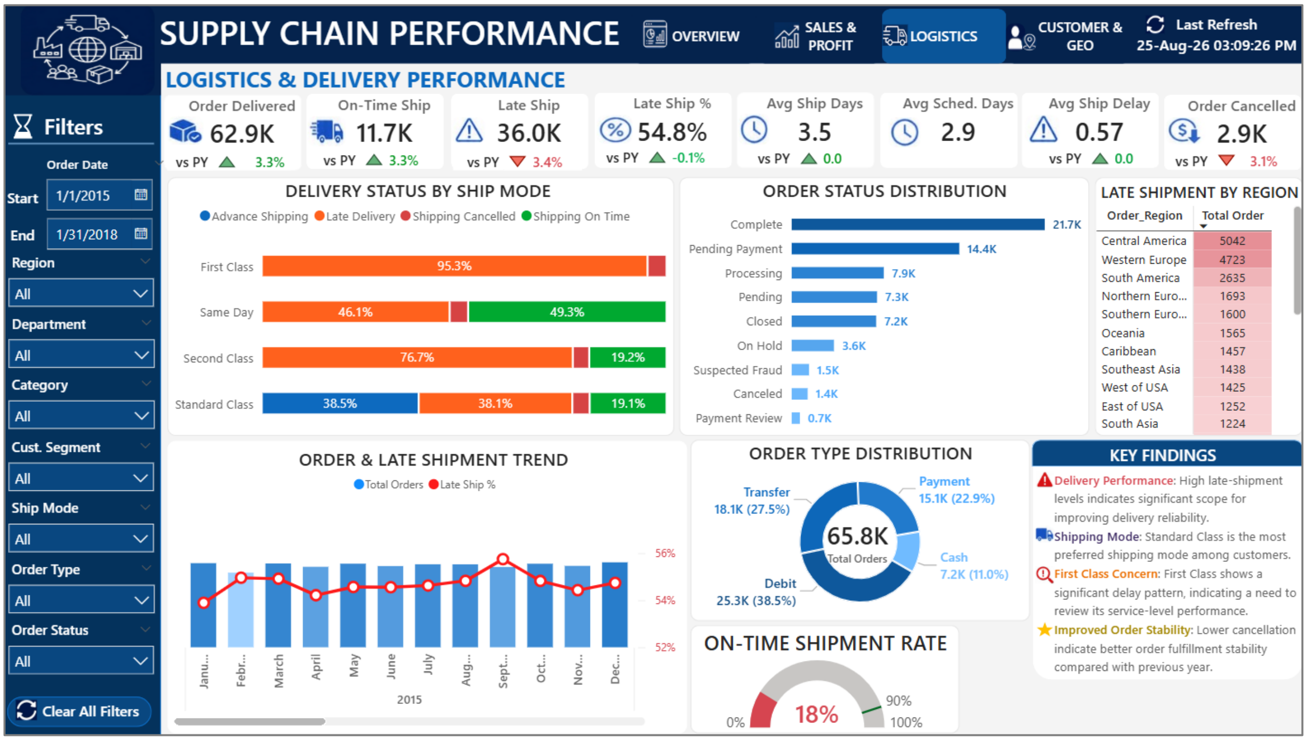


Figure 3. 03_Dashboard_Logistics_Analysis.png — LOGISTICS dashboard page from the supplied project evidence.

The LOGISTICS page converts shipment information into operational performance indicators.

Key analytical views

- Delivered, late, cancelled and on-time shipment KPIs.
- Actual versus scheduled shipping days and average delay.
- Shipment status by shipping mode.
- Order status distribution.
- Late shipments by region.
- Order volume and late-shipment trend.
- Order type distribution and on-time target gauge.

This page supports service-level monitoring and helps identify high-volume shipping modes or regions where late deliveries require operational attention.

- Delivery performance is analyzed both by shipping mode and by geography.
- The 100% stacked delivery-status visual highlights different service profiles by ship mode.
- The gauge provides a direct comparison against an on-time shipment target.
- The late-shipment-by-region table supports operational prioritization.

9.4 Customer & Geography Analysis

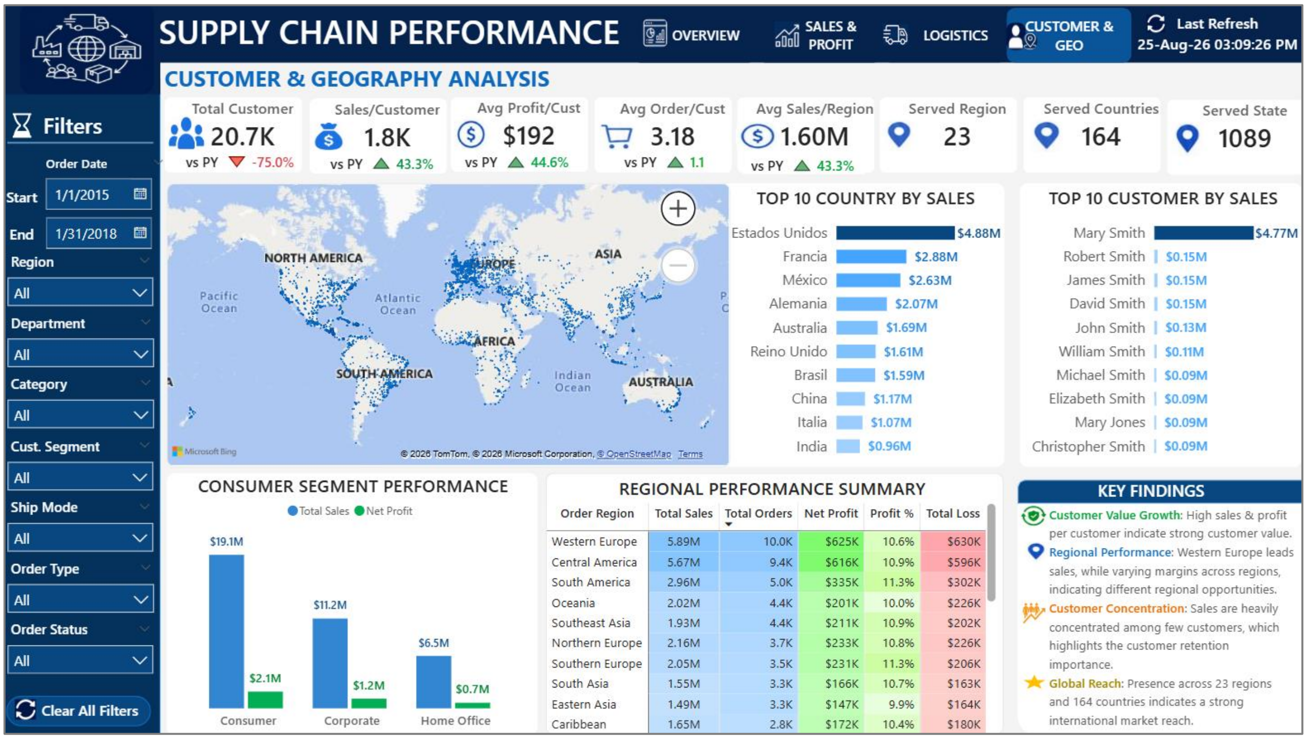


Figure 5. 04_Dashboard_Customer_Geo_Analysis.png — CUSTOMER & GEO dashboard page from the supplied project evidence.

. The screenshot uses a January 2015–January 2018 date selection .

Key analytical views

- Customer count, sales per customer, profit per customer and orders per customer.
- Regional, country and state coverage.
- Top 10 customers and top 10 countries.
- Sales and profit by customer segment.
- Geographic map.
- Regional summary combining sales, orders, profit, margin and loss.

This page is useful for customer prioritization, geographic expansion decisions and concentration monitoring.

- The page connects customer economics with geographic reach.
- Top-10 customer and top-10 country views expose concentration patterns.
- The regional performance summary combines sales, orders, net profit, margin and loss.

10. SQL Layer — Independent Validation Framework

The supplied SQL script is written for MySQL 8.0+ and uses a single analytical table named Supply_Chain_Data. It creates the database/table, imports the cleaned CSV, validates structure and quality, then independently calculates the dashboard's major business metrics.

10.1 SQL query catalog

SQL group	Analysis area	Purpose
Group 01	Database & table setup	Create Supply_Chain_DB and define the Supply_Chain_Data schema.
Group 02	Data import & MySQL configuration	Enable local infile and load the cleaned CSV.
Group 03	Structure & quality validation	Inspect schema/sample rows, count rows, check duplicate order items and missing critical identifiers.
Group 04	Overall business KPI analysis	Calculate sales, profit, margin, customers, orders, AOV, shipping and regional coverage KPIs.
Group 05	Time & trend analysis	Calculate monthly sales, profit and margin.
Group 06	Product & category analysis	Analyze category/product performance and top/loss-making products.
Group 07	Customer segment analysis	Calculate segment sales, profit, customers and sales distribution.
Group 08	Shipping & operational analysis	Compare shipping modes, statuses, scheduled days, actual days and delays.
Group 09	Department performance	Calculate department sales/profit and distribution.
Group 10	Regional performance	Calculate regional orders, sales, units, profit, margin and loss.
Group 11	Customer-level performance	Calculate customer contribution and cumulative sales.
Group 12	YoY validation	Calculate annual performance and prior-year comparisons using LAG().
Group 13	Final data quality validation	Check negative sales, invalid quantity/price/discount/margin and shipping-duration rules.

10.2 Representative validation logic

Overall KPI query — the script independently recalculates the main commercial and service metrics:

```
SELECT
ROUND(SUM(Sales),0) AS Total_Sales,
ROUND(SUM(Order_Profit),0) AS Net_Profit,
ROUND(SUM(Order_Profit)*100/SUM(Sales),2) AS Profit_Margin,
COUNT(DISTINCT Customer_ID) AS Total_Customer,
COUNT(DISTINCT Order_ID) AS Total_Order,
ROUND(SUM(Sales)/COUNT(DISTINCT Order_ID),0) AS Avg_Order_Value,
ROUND(AVG(Actual_Ship_Days),2) AS Avg_Ship_Days
FROM Supply_Chain_Data;
```

The product analysis uses DENSE_RANK() within Category_Name to identify top products by sales and loss-making products. Customer contribution uses window functions to calculate each customer's percentage contribution and cumulative sales percentage. Annual validation uses LAG() to bring prior-year values into the same row.

11. Independent Validation Results

11.1 Main KPI reconciliation

Metric	Cleaned CSV – full dataset	2015–2017 period	Dashboard evidence context
Total Sales	$36.78M	$36.45M	Overview screenshot displays ~$36.5M
Net Profit	$3.97M	$3.93M	Overview screenshot displays ~$3.9M
Profit Margin	10.78%	10.79%	Overview screenshot displays ~10.8%
Distinct Orders	65,752	63,629	Overview screenshot displays ~63.6K
Distinct Customers	20,652	18,529	Customer/GEO screenshot displays ~20.7K under its wider date selection
Average Order Value	$559	$573	Overview screenshot displays ~$573
Average Actual Ship Days	3.50	3.50	Overview screenshot displays ~3.5 days
On-time shipment orders	11,722	11,348	Dashboard shows ~11K–12K depending on date filter

Important interpretation note: dashboard screenshots supplied with the project contain different date selections on different pages. Therefore, the report distinguishes between full-dataset calculations and the 2015–2017 calculations used for the main overview validation rather than treating every screenshot KPI as if it used one identical filter state.

11.2 Independent data-quality results

Validation rule	Rows flagged	Interpretation
Negative Sales	0	No flag
Invalid Quantity (<=0)	0	No flag
Invalid Price (<0)	0	No flag
Invalid Discount (<0 or >1)	0	No flag
Margin flagged (<-100% or >100%)	6,301	Requires semantic review before treating the flag as a data error
Shipping-duration rule flagged	103,400	Requires semantic review before treating the flag as a data error
Missing critical identifiers	0	No flag
Duplicate composite key rows	0	No flag

The zero counts for negative sales, non-positive quantity, negative price, invalid discount range, missing critical identifiers and duplicate composite-key rows support the integrity of the main transaction structure. The margin and shipping-duration rules require interpretation because the underlying business logic can legitimately produce negative profit margins and negative Ship_Delay_Days when actual shipment occurs earlier than scheduled.

11.3 Important SQL interpretation notes

- The SQL data-quality rule treats any negative Ship_Delay_Days as invalid. In the supplied data, negative delay values occur systematically and are consistent with early shipment relative to the scheduled date, so they should not automatically be interpreted as corrupted records.
- The SQL margin-quality rule flags Profit_Margin_Percent below -100%. Loss-making transactions can legitimately have margins below -100% when profit loss exceeds sales value; this is therefore a business-rule review item rather than proof of bad data.

12. Business Findings

12.1 Commercial performance

The full cleaned dataset produces $36.78M in sales and $3.97M in net profit, for an aggregate margin of 10.78%. The 2015–2017 period used by the overview evidence produces $36.45M sales and $3.93M profit.

12.2 Customer segment concentration

Consumer: $19.10M sales, $2.07M profit and 34,11 distinct orders.

Corporate: $11.17M sales, $1.20M profit and 19,85 distinct orders.

Home Office: $6.52M sales, $0.69M profit and 11,777 distinct orders.

12.3 Product and category leadership

Fishing: $6.93M sales and $756K profit.

Cleats: $4.43M sales and $495K profit.

Camping & Hiking: $4.12M sales and $427K profit.

Cardio Equipment: $3.69M sales and $383K profit.

Women's Apparel: $3.15M sales and $350K profit.

12.4 Department concentration

Fan Shop: $17.11M sales and $1834K profit.

Apparel: $7.98M sales and $882K profit.

Golf: $4.61M sales and $498K profit.

Footwear: $4.01M sales and $410K profit.

Outdoors: $1.25M sales and $145K profit.

12.5 Regional performance

Western Europe: $5.89M sales, $625K profit and 10,010 orders.

Central America: $5.67M sales, $616K profit and 9,396.0 orders.

South America: $2.96M sales, $335K profit and 4,979.0 orders.

Northern Europe: $2.16M sales, $233K profit and 3,716.0 orders.

Southern Europe: $2.05M sales, $231K profit and 3,543.0 orders.

12.6 Logistics profile

Standard Class: 39,324 orders, 4.0 average actual ship days and 0.0 average Ship_Delay_Days.

Second Class: 12,778 orders, 3.9 average actual ship days and -1.9 average Ship_Delay_Days.

First Class: 10,079 orders, 2.0 average actual ship days and -1.0 average Ship_Delay_Days.

Same Day: 3,571 orders, 0.48 average actual ship days and -0.48 average Ship_Delay_Days.

12.7 Concentration signal

The largest country by sales is Estados Unidos at $4.88M, followed by Francia at $2.88M. The dashboard's top-country visual therefore provides a useful concentration-monitoring view rather than only a geographic map.

13. Business Recommendations

Recommendation	Action
Protect high-value commercial engines	Prioritize inventory, assortment and promotional planning for the highest-sales categories and Fan Shop, because a large share of commercial value is concentrated in these areas.
Manage discount-driven margin erosion	Use the discount-impact visual to set discount guardrails by product/category. High discount bands should be reviewed for incremental profit rather than sales uplift alone.
Improve delivery reliability	Use ship-mode and regional late-shipment views to identify service combinations with persistent late delivery. Standard Class has the largest order volume, so even small reliability improvements can affect many orders.
Target loss-making products	Combine the product performance table with the SQL top-loss analysis to identify products where sales volume is not translating into healthy profit.
Monitor customer concentration	Use top-customer and segment views to protect high-value accounts while reducing overdependence on a small set of customer names/segments.
Use regional profitability—not only sales	Rank regions using both sales and margin/profit to avoid shifting resources toward high-revenue but comparatively weak-profit markets.
Operationalize independent validation	Run the SQL validation layer after major Power BI refreshes or model changes so that KPI logic remains independently reproducible.
Clarify business rules for negative delay and extreme margins	Document whether negative Ship_Delay_Days represents early shipment and whether margins below -100% are acceptable business outcomes; then align the SQL quality rules with those definitions.

14. Technical Implementation Notes

14.1 Excel / Power Query

- The query is named Cleaned_Data in the supplied image evidence.
- The transformation sequence includes type conversion, text standardization, feature engineering, column cleanup, reordering and duplicate removal.
- The resulting CSV preserves the 43-column analytical schema used by the SQL and Power BI layers.

14.2 Power BI

- The PBIX contains four report sections/pages: OVERVIEW, SALES & PROFIT, LOGISTICS and CUSTOMER & GEO.
- The semantic model includes Fact_Order and six business dimensions plus supporting calculation/utility tables.
- The PBIX contains a dedicated _Measures table and a Last Refresh table.
- The report uses slicers, KPI cards, line/column combinations, bar charts, donut charts, maps, tables, a gauge, navigation buttons and a clear-all-filters action.

14.3 SQL / MySQL

- Target platform stated in the SQL header: MySQL 8.0+.
- The analysis is intentionally single-table; the SQL file states that no JOINs are required.
- Window functions are used for product ranking, customer cumulative contribution and YoY validation.
- The cleaned CSV is imported into Supply_Chain_Data before analytical queries are executed.

14.4 Refresh and reproducibility

The PBIX layout also contains a Last Refresh table and refresh-date card objects. For a production workflow, the refresh timestamp, source path and validation run timestamp should be retained together.

Appendix A — Power BI Measure Catalog

Core commercial

Measure	Business use
Total Sales	Reusable KPI / comparison measure used in the semantic model.
Net Profit	Reusable KPI / comparison measure used in the semantic model.
Profit Margin %	Reusable KPI / comparison measure used in the semantic model.
Total Orders	Reusable KPI / comparison measure used in the semantic model.
Total Customer	Reusable KPI / comparison measure used in the semantic model.
Avg Order Value	Reusable KPI / comparison measure used in the semantic model.

Profitability

Measure	Business use
Total Profitable Order	Reusable KPI / comparison measure used in the semantic model.
Total Loss	Reusable KPI / comparison measure used in the semantic model.
Avg Profit Per Cust	Reusable KPI / comparison measure used in the semantic model.
Avg Profit Per Order	Reusable KPI / comparison measure used in the semantic model.
Avg Discount %	Reusable KPI / comparison measure used in the semantic model.

Logistics

Measure	Business use
Total Order Delivered	Reusable KPI / comparison measure used in the semantic model.
Late Ship Order	Reusable KPI / comparison measure used in the semantic model.
Late Ship %	Reusable KPI / comparison measure used in the semantic model.
Avg Ship Days	Reusable KPI / comparison measure used in the semantic model.
Avg Scheduled Ship Days	Reusable KPI / comparison measure used in the semantic model.
Avg Ship Delay	Reusable KPI / comparison measure used in the semantic model.
Ship Cancelled Order	Reusable KPI / comparison measure used in the semantic model.
On-Time Ship Order	Reusable KPI / comparison measure used in the semantic model.
On Time Ship %	Reusable KPI / comparison measure used in the semantic model.
On-Time Ship Target	Reusable KPI / comparison measure used in the semantic model.

Coverage

Measure	Business use
Served Region	Reusable KPI / comparison measure used in the semantic model.
Served Countries	Reusable KPI / comparison measure used in the semantic model.
Served State	Reusable KPI / comparison measure used in the semantic model.
Sales Per Customer	Reusable KPI / comparison measure used in the semantic model.
Avg Sales Per Region	Reusable KPI / comparison measure used in the semantic model.
Avg Order Count Per Customer	Reusable KPI / comparison measure used in the semantic model.

YoY / comparison

Measure	Business use
YoY Sales %	Reusable KPI / comparison measure used in the semantic model.
YoY Profit %	Reusable KPI / comparison measure used in the semantic model.
YoY Profit Margin Change %	Reusable KPI / comparison measure used in the semantic model.
YoY Orders %	Reusable KPI / comparison measure used in the semantic model.
YoY Customers %	Reusable KPI / comparison measure used in the semantic model.
YoY Avg Order value %	Reusable KPI / comparison measure used in the semantic model.
YoY Avg Profit Per Cust %	Reusable KPI / comparison measure used in the semantic model.
YoY Avg Profit Per Order %	Reusable KPI / comparison measure used in the semantic model.
YoY Avg Ship Days	Reusable KPI / comparison measure used in the semantic model.

YoY Delay Days	Reusable KPI / comparison measure used in the semantic model.
YoY Discount% Change %	Reusable KPI / comparison measure used in the semantic model.
YoY Late Ship % %	Reusable KPI / comparison measure used in the semantic model.
YoY On Time %	Reusable KPI / comparison measure used in the semantic model.
YoY Order Delivered %	Reusable KPI / comparison measure used in the semantic model.
YoY Ship Cancelled %	Reusable KPI / comparison measure used in the semantic model.
YoY Total Loss %	Reusable KPI / comparison measure used in the semantic model.
YoY Total Profitable Orders %	Reusable KPI / comparison measure used in the semantic model.

Appendix B — SQL-to-Dashboard Validation Map

Dashboard concept	Power BI measure / visual concept	SQL validation group
Sales / profit / margin	Total Sales, Net Profit, Profit Margin %	Group 04
Orders / customers / AOV	Total Orders, Total Customer, Avg Order Value	Group 04
Monthly trend	Total Sales, Net Profit, Profit Margin % by Order_Date	Group 05
Category performance	Total Sales, Net Profit, Profit Margin %	Group 06
Product performance / loss	Total Sales, Net Profit, Total Loss	Group 06
Customer segment	Sales, profit and customer distribution	Group 07
Ship mode	Orders, actual days, scheduled days, delay, statuses	Group 08
Department	Sales/profit distribution	Group 09
Region	Orders, sales, units, profit, margin, loss	Group 10
Customer contribution	Customer sales and cumulative contribution	Group 11
YoY	Sales, profit, margin, customer and order comparisons	Group 12
Data quality	Critical IDs, duplicates, numeric and logistics rules	Groups 03 & 13

Appendix C — Scope, Assumptions & Limitations

- This report is grounded in the supplied Excel workbook, cleaned CSV, PBIX package, SQL file and Screenshots evidence.
- The supplied image evidence preserves the Power Query applied-step screenshot and four Power BI dashboard screenshots.
- The exact DAX expression text for every measure is not reproduced here; the PBIX package inspection confirms the measure names and their use in report visuals.
- Dashboard screenshot KPI values can differ from full-dataset calculations because different screenshots use different date selections.
- SQL quality flags are presented as validation-rule outputs and are not automatically classified as data corruption.
- The database source is the Kaggle DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS dataset supplied by the user; the findings are based on the provided cleaned workbook and project artifacts.

Final conclusion

The project demonstrates a strong end-to-end MIS/data-analysis workflow: structured preparation in Excel Power Query, analytical modeling and interactive reporting in Power BI, and independent metric validation in SQL. The resulting solution is not limited to descriptive reporting; it provides a repeatable framework for commercial, customer, geographic and logistics decision-making while preserving a separate validation path for KPI integrity.

15. Detailed Data Preparation Logic

The Excel Power Query layer is the controlled preparation stage between the source data and the analytical model. The supplied Power Query image evidence shows the Cleaned_Data query and the following Applied Steps.

Preparation area	Key applied steps	Purpose
Structure & types	Source; Reordered Columns; Changed Type; Changed Type with Locale	Create a stable analytical table and correctly interpret dates/numbers.
Text standardization	Replace 'and' with &; replace underscores; capitalize words; rename columns	Improve consistency and readability of reporting labels.
Column cleanup	Remove constant/unnecessary columns; Remove Columns2	Keep only fields needed for analysis.
Feature engineering	Ship_Delay_Days; Profit_Category; Customer_Name_Surname; Discount_Range	Create reusable business features before Power BI.
Finalization	Changed Type2; Reordered Columns2; Removed Duplicates	Produce the final clean analytical output.

Key engineered features

- Ship_Delay_Days supports shipment-speed and delay analysis.
- Profit_Category separates profitable and loss-making records.
- Discount_Range enables comparison of sales and profitability across discount bands.
- Customer_Name_Surname creates a cleaner customer display field.
- Removed Duplicates protects the final analytical dataset from duplicate records.

The resulting cleaned dataset contains 180,519 records and 43 analytical fields and is used consistently by the Power BI and SQL layers.

16. Power BI Data Model & Measures

The supplied PBIX separates transactional data from descriptive dimensions and centralizes calculations in a dedicated measure layer.

Model object	Purpose
Fact_Order	Central transaction table containing order-line activity and numeric measures.
Dim_Date	Date filtering and time-based analysis.
Dim_Customer	Customer and segment analysis.
Dim_Product	Product and category analysis.
Dim_Location	Market, region, country, state and city analysis.
Dim_Ship	Shipping-mode analysis.
Dim_Department	Department analysis.
_Measures	Central location for reusable KPIs and YoY calculations.
Last Refresh	Refresh-date reporting.

Core KPI measure groups

Group	Important measures
Commercial	Total Sales, Net Profit, Profit Margin %, Total Orders, Total Customer, Avg Order Value
Profitability	Total Profitable Order, Total Loss, Avg Profit Per Cust, Avg Profit Per Order, Avg Discount %
Logistics	Total Order Delivered, Late Ship Order, Late Ship %, Avg Ship Days, Avg Ship Delay, Ship Cancelled Order, On-Time Ship Order, On Time Ship %
Coverage	Served Region, Served Countries, Served State, Sales Per Customer
YoY	YoY Sales %, YoY Profit %, YoY Orders %, YoY Customers %, YoY Avg Order Value %, YoY Late Ship %, YoY On Time %

This structure allows the same KPI definitions to respond dynamically to dashboard slicers and page filters.

17. SQL Validation & Final Business Insights

SQL validation structure

Validation stage	Purpose
Database & table setup	Create Supply_Chain_DB and Supply_Chain_Data.
CSV import	Load the cleaned dataset into MySQL.
Structure checks	Confirm row volume, schema and sample records.
Data-quality checks	Check duplicates, missing critical fields and invalid numeric/business values.
KPI validation	Recalculate sales, profit, margin, orders, customers, AOV and shipping metrics.
Business analysis	Validate product, category, customer, department, region and shipping results.
YoY analysis	Independently calculate annual comparisons using window functions.

Key findings from the supplied dataset

The full cleaned dataset contains 180,519 records. It covers the period 01 January 2015 to 31 January 2018 and contains 65,752 distinct orders and 20,652 customers.

- Consumer is the largest customer segment by sales.
- Fan Shop is the largest department by sales.
- Fishing is the leading category by sales.
- Standard Class is the highest-volume shipping mode.
- Western Europe and Central America are among the strongest regions by sales.
- Product-level loss analysis and discount-range analysis provide the strongest opportunities for profitability improvement.

Validation interpretation

The SQL layer provides an independent calculation path for Power BI KPIs. The project should compare SQL and Power BI using identical slicer/date context because the supplied dashboard screenshots use different date selections.

The final workflow is therefore: Excel Power Query preparation → Power BI model and dashboard → SQL independent validation → management findings and recommendations.